



Society of Petroleum Engineers

SPE-230081-MS

Strategic Portfolio Reshaping: Applying, Best in Class, Lessons to Unlock ~30% Portfolio Carbon Intensity Reduction

Peter Carydias, Wood, Abu Dhabi, UAE; Vicky Au, Wood, Melbourne, Australia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230081-MS](https://doi.org/10.2118/230081-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Objective (25-75 words): Energy incumbents face risks and opportunities from the energy transition, requiring portfolio-wide decarbonisation (typically >30-40% CI (Carbon Intensity) reduction by 2030) necessitating structural change.

Traditional merger & acquisition evaluation lacks systematic integration of carbon intensity as a core performance metric.

This paper introduces a practical framework to manage portfolio transformation as a key lever for Carbon Intensity reduction, demonstrated through a composite case study for a major NOC targeting ~30% portfolio CO₂e abatement while maximizing commercial value.

Method (75 – 100 Words): Achieving step-change Carbon Intensity reduction via acquisitions and divestments (A&D) requires moving beyond purely financial metrics. Distilling "best in class" project experience and lessons, we present a structured, stage-gated framework embedding Carbon Intensity management throughout the portfolio transformation lifecycle by:

1. Recognising risk through Carbon Intensity targets and baselines to quantify portfolio risk exposure and defining Carbon Intensity-weighted A&D screening criteria.
2. Building value by mandating Carbon Intensity verification and impact modelling within due diligence/valuation, linking Carbon Intensity to near-term efficiency and long-term value drivers (e.g., access to capital).
3. Enabling the vision by implementing governed execution and portfolio-level Carbon Intensity tracking to ensure long-term strategic transformation.

Results (100 – 200 Words):

1. **Risk Management Foundation:** The framework first addresses Carbon Intensity as a strategic risk. Establishing a clear target Carbon Intensity state (e.g., from ~0.45 to ~0.25 tCO₂e/boe by 2030) and mapping the baseline identifies high-Carbon Intensity risk assets. Carbon Intensity-weighted screening criteria explicitly filter targets based on their risk/contribution profile before significant portfolio activity.

2. **Integrating Carbon Intensity for Commercial Value:** Rigorous due diligence quantifies the Carbon Intensity impact of potential opportunities and associated carbon costs/risks. The valuation explicitly incorporates Carbon Intensity factors and carbon pricing (e.g. \$50–70/tCO₂e), demonstrating how lower Carbon Intensity contributes directly to commercial attractiveness (~\$100MM portfolio NPV in the case study). This links decarbonisation to near-term financial evaluation and positions the portfolio for long-term access to debt/capital markets.
3. **Governed Execution for Long-Term Vision:** The framework aligns Carbon Intensity-focused portfolio transformation with corporate investment governance, requiring proposals to articulate Carbon Intensity reduction contributions alongside financial justification. Approved transactions are sequenced onto a transformation roadmap, enabling board-level stewardship beyond short executive tenures. AI-driven portfolio management systems provide ongoing visibility and action nudging, tracking actual Carbon Intensity reduction against the target pathway and verifying the realised financial value, ensuring strategic goals translate into sustained results – a long-term perspective inherent to successful major energy companies.

Novelty (25 – 75 words): This paper presents a practical framework for proactively managing portfolio carbon intensity transformation. By embedding Carbon Intensity as a core risk and value driver within screening, due diligence, valuation, and long-term governance – demonstrated through the NOC composite case study achieving a ~30% abatement pathway with positive NPV – operators can identify and execute the most commercially attractive transactions to meet significant targets, effectively transforming their asset base for sustained value creation in a lower-carbon future.

Extended Abstract / Manuscript

Executive Summary

There are three main levers to achieve carbon intensity reduction ambitions, and we are moving into the domain of the third: Strategic Portfolio Transformation

Energy incumbents face unprecedented portfolio transformation pressures, deploying \$683 billion cumulative (2019-23) into decarbonisation initiatives ([Granskog et al, 2025](#)) - a sixfold increase driven by ambitious targets typically exceeding 30-40% Carbon Intensity reduction by 2030.

To reduce emissions, asset owners are often advised to pursue three key strategies: optimise existing operations, design more sustainable developments, and rebalance portfolios toward lower carbon intensity assets (See [Fig. 1](#)). However, the real challenge frequently lies in realising the value of the strategy through successful implementation.

We previously published our work for successful implementation of the first two strategies for major asset operators from international oil & gas operating companies. The work demonstrated an approach to optimising operations to deliver ~10-20% emissions reduction through discrete programmes targeting rotating reliability, process optimisation, and methane leak reduction ([Carydias & Nielsen, 2023](#)) and sustainable design of a greenfield LNG development to achieve up to 90% CO₂ reduction with an approach adopting localised low-carbon power, monetisation of ramp up gas, process electrification and strategic partnerships ([Carydias & Flowers, 2024](#)). This operational approach also directly addressed methane emissions through systematic leak detection and repair programs, flare reduction initiatives, and process optimisation.

This paper focusses on a method for integrating the third and final component to deliver a comprehensive corporate portfolio strategy for Carbon Intensity reduction – strategic portfolio transformation – and details a composite case study of our application of a three-stage framework to improve the carbon intensity of the portfolios of major international oil & gas companies over time, wherein a ~30% portfolio CO₂e abatement is attainable while maximising commercial value.

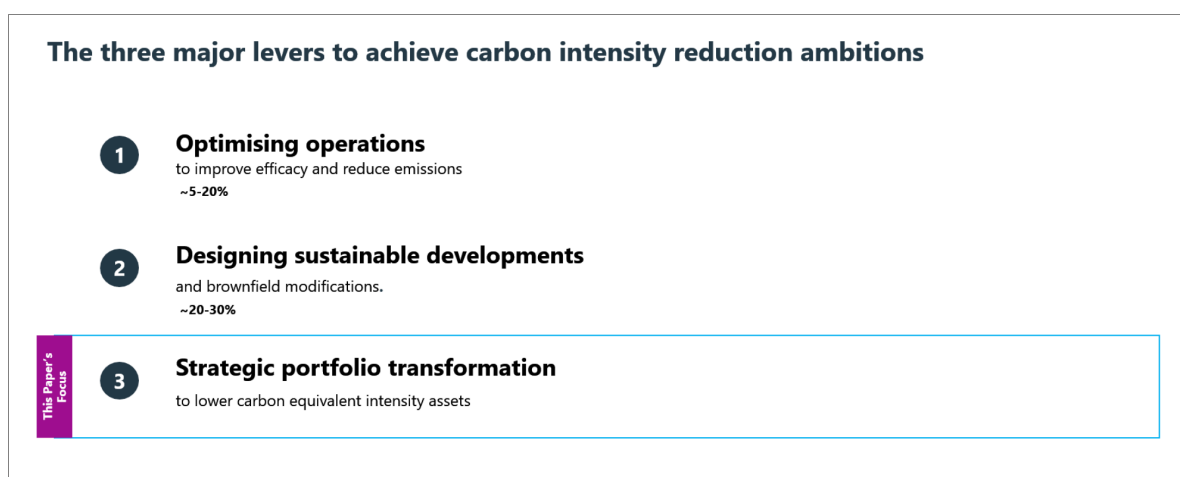


Figure 1—A comprehensive corporate decarbonisation strategy is built upon three distinct levers: optimising operations, sustainable design, and strategic portfolio transformation

A visual representation of application of the portfolio transformation level is depicted in Fig. 2, a combination of targeted divestments of high Carbon Intensity assets and acquisitions or organic growth of low Carbon Intensity assets, with monitoring of this concerted strategy over a longer term horizon to track results against ambitions.

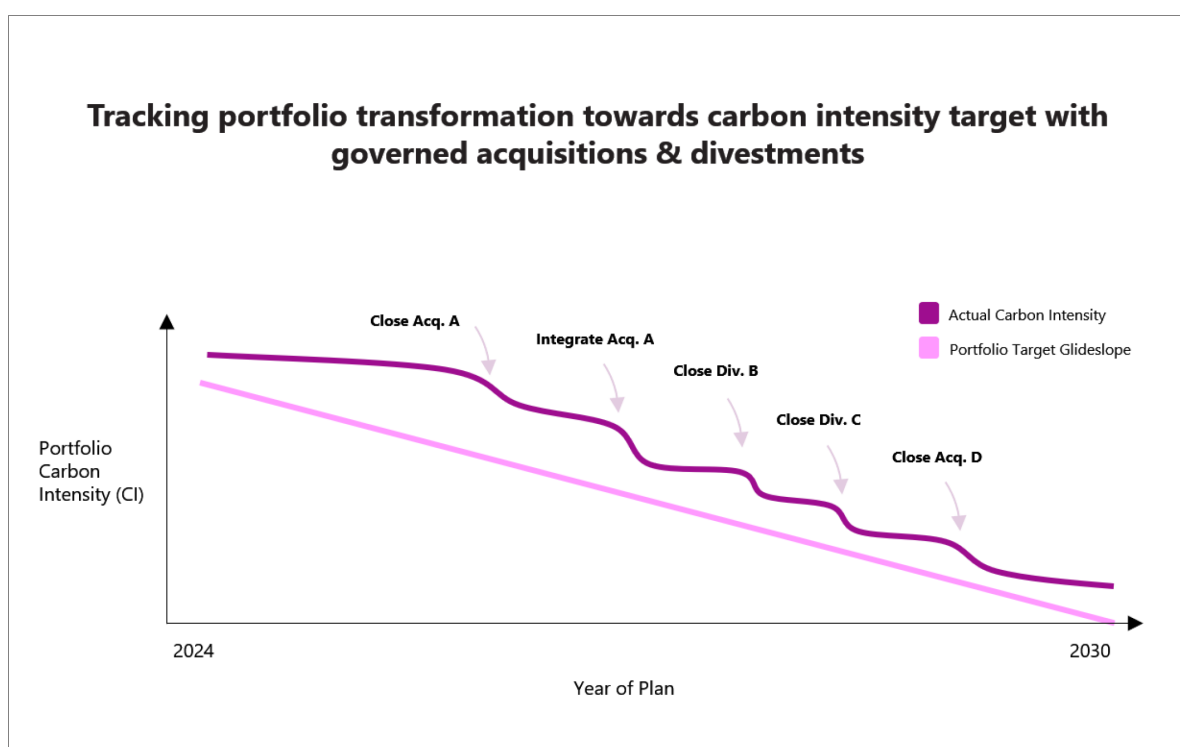


Figure 2—Strategic portfolio transformation achieved by combining targeted divestments of high-Carbon Intensity assets with acquisitions and organic growth of low-Carbon Intensity assets to reshape the portfolio's overall carbon intensity (visual representation)

In our Composite Case study, Major EnergyCo has adopted a phased, integrated advantaged gas strategy to reduce its portfolio carbon intensity with a 2030 target Carbon Intensity of ~0.35 tCO₂e/boe, down from a baseline of 0.45 tCO₂e/boe, with a long-term ambition to reach below 0.15 tCO₂e/boe by 2035.

The Major EnergyCo transformation pathway (Fig. 3) demonstrates how systematic portfolio reshaping bridges the gap between current performance and strategic targets in order to achieve an integrated "advantaged gas" strategy, i.e. low cost, low carbon gas strategy.

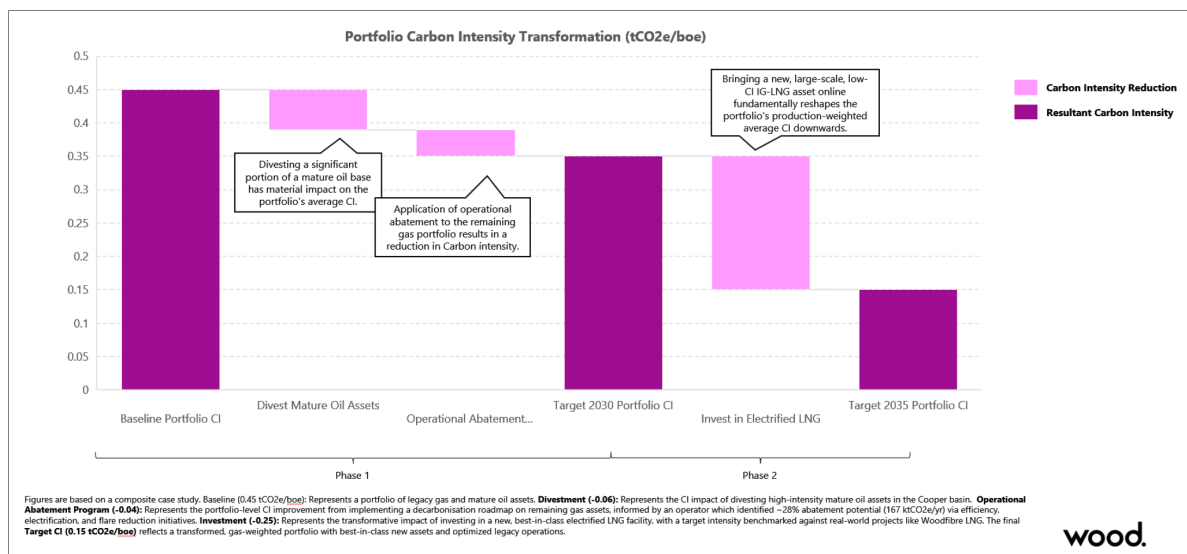


Figure 3—This waterfall chart illustrates how the Composite Case Study producer ("EnergyCo") with a diverse portfolio of legacy gas and mature oil assets reduces its carbon intensity through a multi-lever strategy: divesting high-intensity oil assets, investing in a new best-in-class electrified LNG facility, and implementing an operational abatement program on remaining assets.

Phase 1 (to 2030) focuses on the divestment of high-Carbon Intensity mature oil assets and the application of a comprehensive operational abatement program across remaining legacy gas assets. In the Composite Case Study, this program reflects initiatives with rapid payback periods and is projected to deliver a ~22% intensity reduction by the 2030 target date.

Phase 2 (post-2030) involves the commissioning of a new, best-in-class electrified LNG facility. This major strategic investment provides the step-change required to drive the portfolio's weighted-average Carbon Intensity to its long-term target. The projected carbon price trajectories and positive portfolio Net Present Value (NPV) demonstrate that this phased approach is both environmentally effective and commercially attractive.

How can asset owners implement a practical framework to manage acquisitions & divestitures (A&D) as this critical third decarbonisation pillar, ensuring portfolio transformation achieves Carbon Intensity targets while maximizing commercial value and strategic resilience?

This paper introduces the Accelerate Portfolio Transformation Framework (Fig. 4) - a structured, three-stage methodology embedding Carbon Intensity management throughout the A&D lifecycle.

The three stages are:

1. **recognising risk** through Carbon Intensity targets and baselines to quantify portfolio risk exposure and defining Carbon Intensity-weighted A&D screening criteria.
2. **building value** by mandating Carbon Intensity verification and impact modelling within due diligence/valuation, linking Carbon Intensity to near-term efficiency and long-term value drivers (e.g., access to capital).
3. **enabling the vision** by implementing governed execution and portfolio-level Carbon Intensity tracking to ensure long-term strategic transformation.

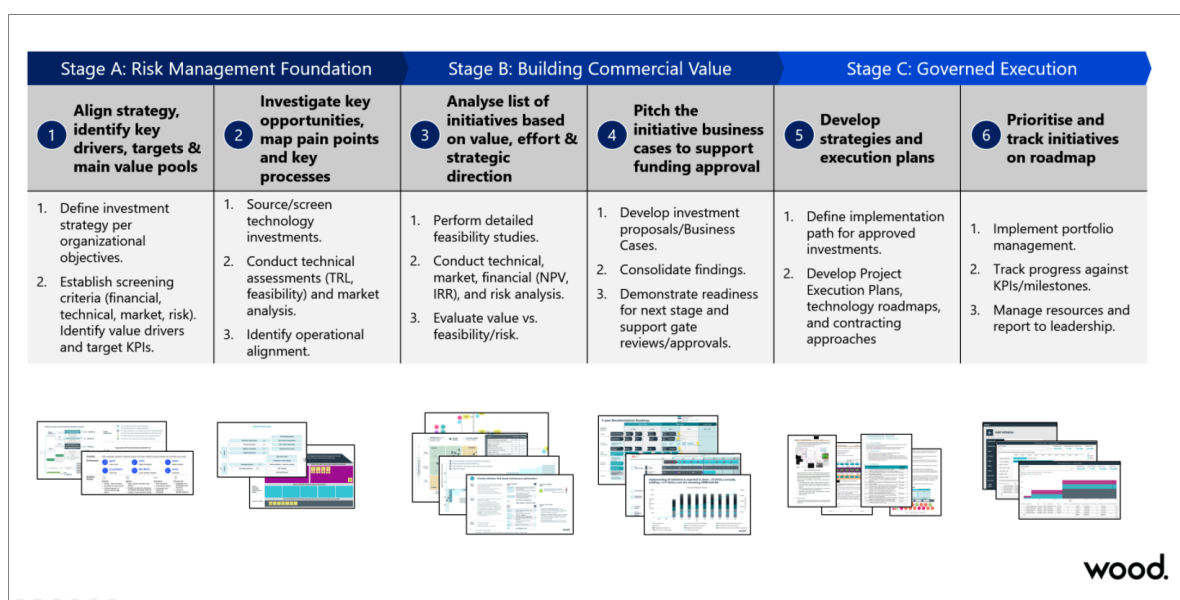


Figure 4—The Accelerate Portfolio Transformation Framework - a structured, three-stage methodology systematically embedding Carbon Intensity management throughout the A&D lifecycle

The framework is adaptable and has been applied successfully to major oil & gas operations at an asset level to identify, prioritise, and monitor initiatives. This includes developing portfolio roadmaps targeting a ~28% (~170 ktCO₂e/yr) reduction in Scope 1 & 2 emissions for existing assets, while simultaneously guiding greenfield projects like LNG facilities toward best-in-class carbon intensity targets of ~0.1 tCO₂e/tLNG or lower - a significant improvement on the regional average of ~0.32 tCO₂e/tLNG. For this discussion, acquisitions and divestitures can refer to acquisitions (or divestments) of a corporate entity, asset or operation, or technologies.

Table 1 provides a consolidated overview of this framework in the context of the Major EnergyCo composite case study, demonstrating its application and the value generated at each stage. The framework synthesises "best in class" lessons from operational optimisation and sustainable design to enable strategic portfolio reshaping. By integrating quantitative risk assessment methodologies, comprehensive due diligence frameworks, and governed execution systems, companies can execute their chosen strategy while delivering commercially attractive decarbonisation pathways that position them advantageously in an evolving energy landscape.

Table 1—A stage-by-stage summary of the Accelerate Portfolio Transformation Framework, detailing the objective, activities, and value demonstrated through the Major EnergyCo composite case study.

Accelerate Portfolio Transformation Framework Stage	Accelerate Portfolio Transformation Framework Step	Case Study Application & Value Demonstration (Composite Case Study "Major EnergyCo")	Typical Artefacts Delivered per Step
Stage A: Risk Management Foundation	1: Align Strategy, Identify Drivers, Targets & Value Pools	Objective: Define a clear pathway for Major EnergyCo to achieve a mandated >30% portfolio Carbon Intensity reduction by 2030. Activities: Facilitated strategic alignment workshops, establishing a clear 'Roadmap Basis' Defined strategic drivers, quantifiable targets (35% intensity reduction vs. 2018 baseline), scope (Scope 1 & 2 operated assets), and explicit investment criteria (positive NPV - net present value including \$50-70/tCO₂e internal carbon price, <5yr payback). Established a verified portfolio Carbon Intensity baseline (~600 ktCO₂e/yr), benchmarking key assets. Value Delivered in Composite Case Study: Established unambiguous targets and a data-driven baseline, focusing	Roadmap Basis Structure Baselining Methodology Benchmarking Concept Criteria Definition

Accelerate Portfolio Transformation Framework Stage	Accelerate Portfolio Transformation Framework Step	Case Study Application & Value Demonstration (Composite Case Study "Major EnergyCo")	Typical Artefacts Delivered per Step
Stage B: Building Commercial Value		subsequent efforts. Saved screening time by defining clear Carbon Intensity/financial hurdles upfront.	
	2: Investigate Opportunities, Map Pain Points & Key Processes	Objective: Identify and screen potential A&D and major internal decarbonisation projects. Activities: Conducted portfolio-wide screening (>130 opportunities identified) via CoLab (Collaborative Innovation) workshops & SME input. Applied Carbon Intensity-weighted screening criteria resulting in 50 shortlisted opportunities. Performed initial technical (TRL - Technology Readiness Level) and market landscape assessments. Value Delivered in Composite Case Study: Generated comprehensive opportunity list. Rapidly filtered >60% of initial ideas, focusing detailed analysis resources (due diligence budget saved) on most promising candidates.	Opportunity Sourcing/ Ideation Emissions Source Mapping: Screening Criteria Initial Assessment / Tech-Market Scan
	3: Analyse List of Initiatives (Deals) Based on Value, Effort & Strategy	Objective: Perform rigorous techno-economic evaluation for objective prioritisation. Activities: Executed detailed feasibility studies. Developed MACC - Marginal Abatement Cost Curve analysis integrating CAPEX/OPEX and abatement potential. Performed LCA – lifecycle analysis /Carbon Intensity validation and detailed risk assessment. Value Delivered in Composite Case Study: Quantified total abatement potential (~170 ktCO₂e/yr, ~28%). MACC revealed ~74% initiatives had negative abatement costs. Identified specific high-value levers (e.g., Compressor Electrification saving ~7M/yr revenue, 17M/yr revenue, 750k/yr OPEX, abating ~82 ktCO₂e/yr).	MACC Methodology & Visual Initiative Quantification LCA/Carbon Intensity Methodology Risk Assessment
Stage C: Governed Execution	4: Pitch Initiative Business Cases for Funding Approval	Objective: Secure board approval and funding for the optimised decarbonisation roadmap. Activities: Developed standardised 1-page business cases for top ~23 initiatives. Synthesized findings into a portfolio recommendation highlighting optimal pathway (achieving ~28% target with 86MM NPV)	Initiative Summary Template Prioritisation Visuals Portfolio Value Summary Governance Process Alignment
	5: Develop Strategies & Execution Plans	Objective: Translate approved initiatives into actionable implementation plans. Activities: Developed detailed implementation strategies, PEP - Project Execution Plans (scope, schedule, resources, contracting), and integration/separation playbooks for the top initiatives. Value Delivered in Composite Case Study: Provided clear execution blueprints. Established clear baselines for project control.	PEP Development Integration/Separation Planning
	6: Prioritise & Track Initiatives on Roadmap	Objective: Ensure portfolio value delivery through structured tracking and governance. Activities: Established PMO - Portfolio Management Office support. Implemented visual roadmaps sequencing the 23 initiatives. Deployed portfolio dashboards tracking Carbon Intensity abatement (+2400 ktCO₂e by 2038 forecast for another operator) & financial performance (\$86M NPV forecast). Quarterly reporting. Value Delivered in Composite Case Study: Established portfolio visibility. Enabled proactive risk/resource management. Provided auditable tracking of progress towards Carbon Intensity target & financial benefits.	Roadmap Visualisation Portfolio Tracking Dashboards: PMO/Governance Reporting: -

Method

A Structured Three-Stage Framework: Embedding Carbon Intensity Throughout the A&D Lifecycle

We now outline the method and its application in detail with reference to Major EnergyCo as the composite case study.

Stage A: Setting the Risk Management Foundation and enabling operators to focus resources on opportunities offering the best overall return

Right or wrongly, emissions reduction today is largely seen as a cost driver for asset owners, not a value driver. We see first-hand the consistent challenge this presents in corporate functions attempting to implement emissions reduction strategies and initiatives with long term targets in the face of near-term value-driving key performance indicators (KPIs) at the asset level such as production and operational expense (OPEX). However, consistent stalling or delaying of emissions reduction ultimately translates to a real risk in an increasingly carbon-conscious market. Corporate strategy must balance near-term focus with the long-term relevance of the portfolio. Establishing risk management as a foundation of the three-stage framework represents our collated project experience of organisations best set up for success – those that recognise, quantify and mitigate the risk exposure presented by Carbon Intensity on a portfolio basis by defining Carbon Intensity targets and baselines and Carbon Intensity-weighted A&D screening criteria.

Align Strategy, Identify Drivers, Targets & Value Pools. Step 1 (Fig. 4) establishes the strategic foundation and investment criteria to unlock transformation value by defining unambiguous Carbon Intensity targets, quantifying data-driven baselines, and setting explicit financial hurdles to focus subsequent efforts. The aim is to establish clear Carbon Intensity / financial criteria upfront, enabling efficient resource allocation toward genuine transformation opportunities while avoiding the trap and time-consumption associated with pursuing initiatives that fail to meet strategic decarbonisation or commercial thresholds. Steps 1a-1c provide more detail.

Strategic Archetype Selection and Competitive Positioning. The framework begins by establishing clear positioning within a handful of strategic archetypes, as each requires fundamentally different portfolio transformation approaches and operating model adaptations. A **strategic archetype** refers to a fundamental pattern or template that characterises how organizations approach strategy formulation and implementation. These archetypes represent recurring configurations of strategic behaviours, decision-making processes, and organisational structures that companies tend to adopt. The relevance of archetype identification is in the understanding and appreciation of one's strategy and ambition as matched to strengths, capabilities and motivations which in turn guides decisions on investments and divestments. **Resource specialists** may focus on high-grading portfolios toward advantaged hydrocarbon growth opportunities, concentrating investments on resources offering the best combination of lower break-even prices and lower emissions intensity. **Integrated energy players** may seek to retain profitable core hydrocarbon assets while capturing opportunities in rapidly growing low-carbon markets including renewable power, bioenergy, next-generation mobility, and hydrogen, betting they can emerge as natural owners based on incumbent capabilities and relationships. **Low-carbon pure plays** take this thinking further, divesting legacy high-carbon portfolios while building future-proof low-carbon businesses, like Ørsted's transformation to become the world's largest offshore wind developer.

Portfolio Baseline and Advantaged Asset Identification. Next, we establish verified, granular portfolio baselines while identifying "advantaged" assets across four key dimensions, such as: resilience (low development costs, appropriate fiscal terms, high margins), sustainability (low emissions, high throughput, access to clean electricity), predictability (subsurface knowledge, stable conditions, government support), and timeliness (market access, partner alignment, short cycle times).

We established a baseline of ~600 ktCO₂e/yr for Major EnergyCo, enabling precise identification of high-intensity divestiture candidates and low-Carbon Intensity acquisition targets. This process incorporates

primary emission-intensity drivers such as resource complexity (with API gravity $\leq 20^\circ$ assets averaging 3x more emission-intensive than $\geq 50^\circ$ assets), processes and engineering choices (with hub platforms requiring more energy than simple installations), and routine flaring/venting prevalence (contributing up to 40% of regional carbon intensity).

Value Pool Quantification and Investment Threshold Setting. We then establish explicit, multi-criteria screening approaches considering the substantial value pools available from portfolio transformation. For example, as of 2019, there was \$40 billion annual revenue potential from monetising currently wasted gas (257 bcm globally equivalent to nearly half of Europe's consumption) and up to 20% energy savings opportunities representing \$10 billion annual industry value, and the growing climate technology.

We deployed investment criteria that explicitly weighted Carbon Intensity Impact, Financial Return, and Strategic Fit, ensuring A&D pipelines support strategic transformation while maintaining commercial discipline. This saved Major EnergyCo screening time by defining clear Carbon Intensity/financial hurdles upfront, focusing subsequent efforts on opportunities offering genuine transformation potential aligned with their chosen integrated energy player archetype.

Filter by Investigating Opportunities, Mapping Pain Points & Key Processes. Step 2 casts a wide net to comprehensively identify transformation opportunities before applying rigorous Carbon Intensity-weighted criteria to rapidly filter prospects, preventing analysis paralysis from evaluating too many marginal candidates. This screening and filtering process is visualised in Fig. 5.

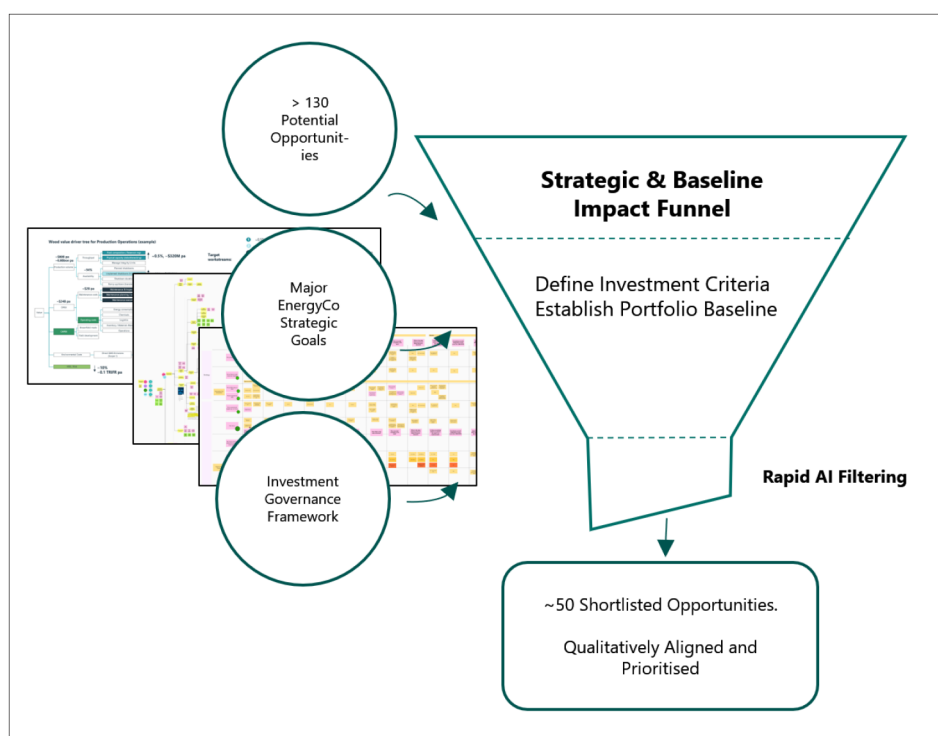


Figure 5—Systematic Screening Aligns Opportunity Pipeline with Strategy & Baseline

In the Major EnergyCo case, our portfolio-wide screening identified >130 opportunities on which rapid AI filtering was applied to provide ~50 promising candidates, generating comprehensive opportunity coverage while saving substantial due diligence resources by focusing detailed analysis only on initiatives offering optimal Carbon Intensity reduction and commercial potential. We have found that performing this prioritisation using a matrix that plots opportunities based on Carbon Intensity impact versus strategic fit, as shown in Fig. 6, is effective for identifying the highest-value candidates. Steps 2a-2b provides the detail.



Figure 6—Prioritising A&D Opportunities by Financial Value & Carbon Intensity Impact

Opportunity Identification and Market Assessment. We deployed systematic workshop methodologies and subject matter expert input to conduct portfolio-wide screening, building on approaches that have successfully identified opportunities across diverse energy contexts. We identified >130 opportunities for Major EnergyCo through this approach, incorporating initial technical assessments and market landscape evaluations to screen substantive opportunities to detailed evaluation. This market context is included so that opportunity identification considers not just current technology availability but projected scaling trajectories and cost reduction curves essential for long-term strategic positioning as part of risk exposure management.

Rapid Strategic Filtering and Resource Optimisation. To save significant due diligence budget while ensuring concentrated analytical effort on opportunities offering optimal Carbon Intensity reduction and commercial potential, we implement rapid filtering of opportunities with screening criteria. Without rapid screening criteria, companies risk getting overwhelmed by extensive opportunity lists and losing focus on value-creating initiatives. Major EnergyCo's Carbon Intensity-weighted screening criteria enabled rapid filtering of >60% of initial opportunities (from >130 to ~50 candidates), demonstrating the critical importance of systematic criteria application to avoid resource dissipation across marginal prospects. The process described in this document, aided by AI, reduced this screening process from weeks to ~2 days.

The filtering process is based on the changing risk-reward trade-offs characterised by [Savenkova \(2025\)](#) who showed that decision-making has evolved from a simple two-pillar model of profitability and technical risk to a complex, multi-dimensional framework. This new framework incorporates a third pillar of sustainability (including emissions and carbon price). This economic reality drives our methodology's emphasis on rapid identification of opportunities offering both Carbon Intensity reduction and enhanced

returns, particularly important as companies navigate the capital allocation dilemma between near-term reinvestment and next-decade strategic positioning.

Stage B: Building Commercial Value, and demonstrating to Operations how carbon intensity improvements directly correlate with operational efficiency gains, energy cost reductions, and equipment reliability

Having recognised and accepted the risk exposure presented by Carbon Intensity and put in place effective steps to manage, the next task is to shift from "how do I mitigate this risk" to "how do I drive value" and embed this into the strategic mindset and methodology. Stage B's focus on building commercial value into Carbon Intensity decisions integrates the definition of value from the standpoint of different stakeholders, linking Carbon Intensity to both immediate incentives and near-term impact such as lower operational costs, reliability and greater operational efficiency (typically the drivers at the asset level), and longer-term benefits such as future market access in a carbon-conscious world, access to capital from carbon-conscious lenders to fund growth, and other "future-proofing" strategies (typically the concern at the corporate level), thus bridging the gap that we have found is a frequent failure point in strategy implementation and transformation initiatives. Steps 3 and 4 below provide the detail, illustrated through the facts and figures of Major EnergyCo.

Analyse List of Initiatives (Deals) Based on Value Potential, Effort & Strategy. Step 3 focuses on tying together and defining both the environmental and financial value to rank portfolio decisions, driving a funnel of quantified transformation value potential through techno-economic evaluation to reveal the commercial attractiveness of decarbonisation initiatives. Fig. 7 illustrates this from the Major EnergyCo case with Marginal Abatement Cost Curve (MACC) analysis demonstrating that some opportunities offered negative abatement costs, representing immediate value accretion, and many were attractive in the current carbon price environment.

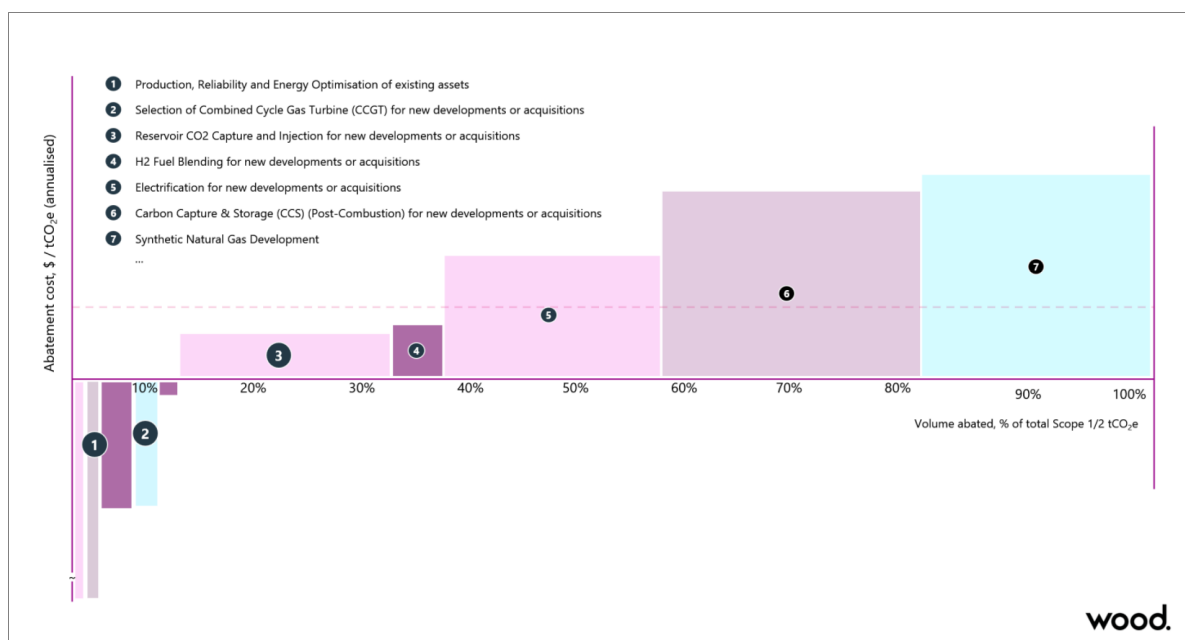


Figure 7—Marginal Abatement Cost Curve (MACC) demonstrating commercial attractiveness of decarbonisation initiatives

Steps 3a and 3b provide more detail behind the process as applied to Major EnergyCo.

Risk-Based Technical Due Diligence and Performance Benchmarking. The framework first establishes risk-weighted criteria to determine appropriate due diligence depth across four levels (Basic, Intermediate,

Advanced, Detailed) based on factors including asset complexity; operator track record, technical risk exposure, and investment scale.

This systematic risk assessment then guides comprehensive technical evaluation encompassing engineering assessment, safety systems evaluation, environmental due diligence, and techno-economic validation. The process produces integrated MACC analysis enabling direct comparison of A&D opportunities with internal optimisation projects across the three-pillar decarbonisation strategy.

Carbon-Adjusted Valuation and Strategic Value Creation. The evaluation of Major EnergyCo's quantified near-term abatement potential (~170 ktCO₂e/yr, ~28% portfolio reduction) with MACC analysis revealed ~74% of initiatives demonstrated negative abatement costs, indicating immediate value accretion opportunities. Specific high-value levers identified include Compressor Electrification generating \$7 million/year OPEX savings, \$17 million/year revenue enhancement, and \$750,000/year additional operational benefits while abating ~82 ktCO₂e/yr. This multi-pronged approach builds on the understanding that an increase in production efficiency delivers a reduction in emission intensity (~10% uplift resulting in 4% reduction of emissions intensity), enabling quantification of both operational and financial benefits from portfolio optimisation.

Valuation explicitly incorporates Carbon Intensity factors and carbon pricing (\$50-70/tCO₂e in the case study), demonstrating how lower Carbon Intensity contributes ~\$100 million portfolio NPV impact. This approach addresses the fundamental economic reality that breakeven economics of many high-emission-intensity reservoirs are marginal above \$65, making disciplined portfolio transformation both an emissions reduction and financial optimisation strategy.

Pitch Initiative Business Case on a page for Go-/No-Go (Funding) Approval. Step 4 produces rapid go/no-go Investment Committee decisions through standardized one-page business cases that synthesise complex technical analysis into clear portfolio recommendations highlighting the optimal transformation pathway. This streamlined approach accelerates decision-making by distilling detailed feasibility work into actionable investment propositions, demonstrating how selected initiatives collectively achieve Carbon Intensity targets while delivering positive portfolio NPV (\$86M in the case study), encouraging timely capital allocation and executive commitment. Steps 4a to 4d outlines this part of the methodology in further detail.

Rapid Business Case Development and Portfolio Communication. The framework employs templates to develop standardised one-page business cases for top opportunities (~23 in Major EnergyCo case study), synthesising findings into portfolio recommendations highlighting optimal pathways to achieve the portfolio carbon intensity targets. While different organisations will adapt their own business case template, the crux of this communication is it needs to address the balance between legacy core hydrocarbon businesses with diversification of the operating model into low-carbon options, which may require new entrepreneurial capabilities and cultures.

Business cases explicitly articulate Carbon Intensity reduction contributions alongside traditional financial justification, recognising the investors and stakeholders looking to sustainable returns and capital discipline over pure growth investments. The synthesis enables clear communication of both individual initiative merits and portfolio-level transformation benefits to communicate the integrated nature of strategic portfolio reshaping to decision-makers.

Stakeholder Alignment and Investment Return Expectations. Portfolio transformation initiatives must navigate competing stakeholder expectations, particularly the tension between shareholders/investors and debt providers who operate with fundamentally different time horizons and decarbonisation priorities.

Our survey of challenges faced by emissions strategy owners found that the increasing demand of equity investors for both near-term returns and long-term sustainability positioning creates pressure for transformation initiatives to deliver immediate value while building future-resilient portfolios. An

appreciation of the variance in investor/ stakeholder time horizons is needed - activist investors may prioritise shorter-term quarterly performance while institutional investors might focus on a longer-term three- to five-year value creation, complicating capital allocation decisions for multi-year transformation programs.

An additional potential category of stakeholder - debt providers - typically operate with longer-term risk frameworks, incorporating ESG (environmental, social and governance) criteria and stranded asset risks into lending decisions. Banks and institutional lenders may offer preferential terms for lower-carbon portfolios while tightening conditions for high-emission assets, creating direct financial incentives for strategic transformation beyond equity market pressures.

The framework acknowledges the feedback from these challenges and addresses these tensions by demonstrating how carbon-adjusted valuations serve both constituencies - showing equity investors the NPV impact of transformation initiatives while providing lenders with risk management and improved debt service capacity through operational efficiency gains. Clear communication of both near-term financial benefits (OPEX savings, revenue enhancement) and long-term positioning advantages (capital access, market resilience) enables alignment across stakeholder groups with different investment realization timeframes and decarbonization motivations.

Investment Committee Governance and Strategic Alignment. Step 4 builds alignment with corporate investment governance through integration with formal capital management systems, requiring Investment Committee gate reviews before proceeding to implementation. The process requires explicit articulation of how proposed transactions support the chosen strategic archetype while meeting established financial hurdles including carbon-adjusted NPV thresholds. This governance alignment reduces the pursuit of commercially attractive but strategically misaligned assets while allowing for the evaluation of long-term competitive positioning alongside near-term financial metrics.

Stage C: Establishing Governed Execution and providing operational businesses with clear KPIs linking emissions performance to cost optimisation and production efficiency

To preserve the efforts of risk management and value association for Carbon Intensity improvement of Stages A and B, in Stage C we apply a principle derived from the third "best practice" finding of our executive survey: the recognition that successful transformation requires "long-termism", and that board-level and/or shareholder commitment extending beyond individual executive tenures is needed to ensure strategic persistence across market cycles and leadership changes. This is particularly relevant amidst a global trend in declining average chief executive tenure, reaching a new low in Q1 of 2025 of 6.8 years according to the Global CEO Turnover Index. Against the backdrop of 2040, 2050 and 2060 emissions reduction, neutrality or carbon negative targets, the probable number of changes in executive leadership over that time strengthens the need to establish execution procedures for stable governance and a long-term view to value realisation.

Develop Strategies & Execution Plans. The first part to Stage C is concerned with value execution through the development of implementation blueprints and clear project control baselines to reduce the planning cycle while providing coordinated delivery across diverse transformation initiatives. This translates strategic intent into actionable project execution plans, integration playbooks, and partnership frameworks essential for successful transformation delivery, establishing the operational foundation required to realise both decarbonisation targets and commercial value from approved initiatives. The process is detailed in Steps 5a to 5c.

Comprehensive Project Execution Planning and Integration Management. In Step 5a, we develop detailed Project Execution Plans (PEPs) covering scope, schedule, resources, and contracting strategies for approved initiatives (~23 in Major EnergyCo case study), building on proven project execution methodologies from major infrastructure implementations. Integration and separation playbooks provide

approaches for both acquiring low-Carbon Intensity assets and divesting high-Carbon Intensity positions, incorporating specialised knowledge from complex energy project management. The comprehensive planning approach from major engineering project execution principles and experience insights reduced Major EnergyCo's planning cycle while establishing clear baselines for project control across the portfolio transformation initiative.

The specific challenge we address with this part of the methodology is that portfolio transformation success depends on coordinated execution across diverse project types spanning operational optimisation (first pillar), sustainable design (second pillar), and strategic A&D (third pillar), and requires sophisticated program management capabilities to ensure synergies are captured and conflicts avoided.

External Partnership Requirements and Capability Integration. Portfolio transformation initiatives inherently involve complex external interfaces requiring specialised capabilities that typically exceed internal organisational capacity. To address this, in Step 5b we conduct internal resource and capability audits to identify gaps, then develop strategic partnerships to address these requirements across both evaluation and implementation phases.

During due diligence, external advisory support proves essential for carbon market navigation, regulatory compliance assessment, and specialised technical evaluation of low-carbon technologies. For implementation, partnerships become critical for technologies like carbon capture and storage (demonstrated by projects like Northern Lights requiring industry collaboration) and renewable energy integration demanding specialised firming infrastructure capabilities.

Each partnership addresses specific capability gaps while managing the economic and political complexities inherent in transformation initiatives so that companies can execute strategic repositioning without developing all required expertise internally.

Organisational Integration and Change. It is a critical reality that transformation programmes often fail due to organisational integration challenges. Change management is needed to prevent common failure modes that derail even technically sound initiatives.

In Step 5c we establish measures to address the three common failure points of transformation programmes: Failure to Launch, Failure to Sustain, and Failure to Scale using findings from our client survey and collated project experience. The framework recognises that while 90% of known technological solutions cost less than \$50/tCO₂e, successful implementation requires appropriate organisational wiring to achieve projected performance rather than just technical capability.

In our work on Major EnergyCo, organisational integration across ~23 initiatives helps to ensure individual projects collectively deliver portfolio-level Carbon Intensity targets through coordinated change management, clear accountability structures, and integrated performance tracking, avoiding the common pitfall where technically successful projects fail to achieve strategic transformation objectives due to poor organizational coordination and change management execution. Generally, this may take the form of quarterly Carbon Intensity performance scorecards, dedicated strategic sessions on archetype execution and annual portfolio stress-testing under different carbon pricing scenarios, supported by director education on evolving carbon markets and decarbonisation technologies.

Prioritise & Track Initiatives on Roadmap: Portfolio Governance and Long-Term Value Realisation.

Step 6 supports long-term transformation success through dedicated governance structures and sustained portfolio tracking that transcend short-term market cycles and executive tenure changes, recognising that meaningful portfolio transformation requires multi-year commitment and institutional persistence. This stage establishes visual roadmaps, PMO oversight, and integrated performance dashboards tracking Carbon Intensity progress and financial delivery across extended timelines, as consolidated in the example in Fig. 8, ensuring transformation initiatives receive the long-term focus and resource commitment necessary to

compete effectively against shorter-cycle operational priorities and deliver sustained strategic advantage. Steps 6a to 6c provide the detail to this process.

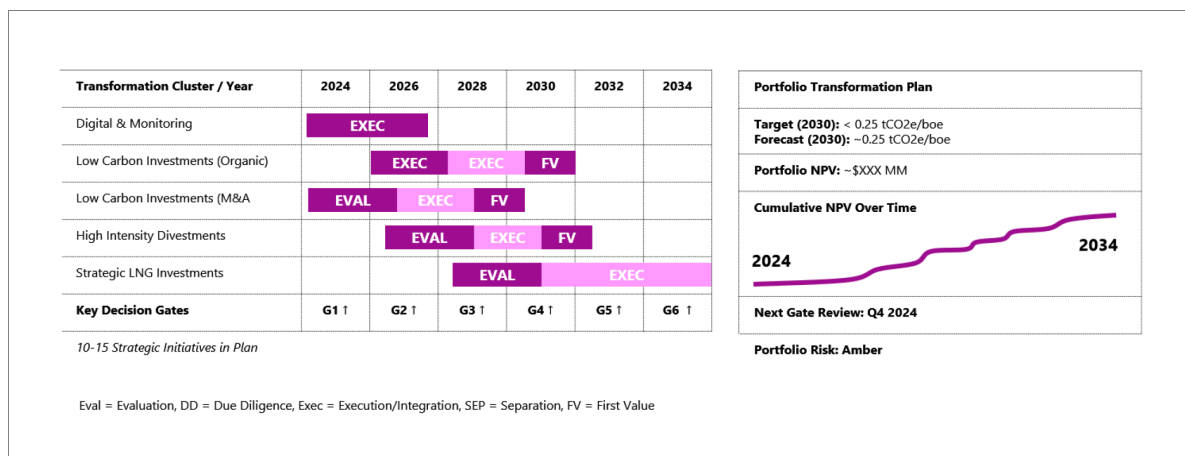


Figure 8—Project roadmap and performance tracking dashboard establishing portfolio-level tracking to monitor progress against key financial and decarbonisation metrics.

Visual Roadmap Implementation and Portfolio Management Office. In Step 6a, we develop a programme on a page and, depending on existing structures, dedicated Project Management Office (PMO) support to provide oversight extending beyond individual tenures. PMO governance incorporates proven reporting methodologies while building on long-term management agreement experience. From a best practice perspective, quarterly reporting cycles maintain stakeholder engagement while enabling proactive risk and resource management across the transformation portfolio so that strategic momentum is maintained across market cycles and competitive pressures. This provides board-level visibility into both progress against Carbon Intensity targets and financial value delivery, essential for maintaining shareholder and lender confidence during multi-year transformation periods.

Digital Portfolio Tracking and Performance Governance. In Step 6b, digital portfolio tracking systems enable governance bodies to monitor transformation progress over time, providing visibility into both Carbon Intensity performance and financial value delivery essential for sustained strategic oversight. Portfolio dashboards track Carbon Intensity abatement and cumulative value achieved (e.g. \$86 million NPV delivery in the composite case study) through integrated governance platforms enabling informed decision-making and course corrections when needed.

We deploy AI-driven portfolio management systems here to provide sophisticated orchestration through integrated technological components: AI-powered "control towers" unify data from financial tracking, operational performance, and emissions monitoring systems, with the potential for real-time visibility across transformation roadmaps; predictive analytics engines trained on historical project data identify potential risks before they materialise, enabling proactive intervention when initiatives deviate from planned trajectories; digital portfolio models integrate and validate data from all transformation sources to reveal critical interdependencies; and dynamic resource optimisation algorithms continuously rebalance resources, schedules, and priorities based on real-time constraints, ensuring the organisation remains aligned to achieve strategic transformation objectives while adapting to evolving market conditions and stakeholder requirements.

Digital tracking provides board-level visibility into portfolio transformation effectiveness while enabling proactive intervention when initiatives deviate from planned trajectories. We have incorporated our digital capabilities here recognising that systematic data collection and performance visualisation are essential for maintaining stakeholder confidence and ensuring accountability across multi-year

transformation timelines. Digital governance systems provide ongoing performance transparency and trend analysis, enabling leadership to demonstrate progress against strategic commitments while identifying optimisation opportunities that maximize both Carbon Intensity reduction and commercial value across the transformation portfolio.

Long-Term Value Delivery and Strategic Advantage Maintenance. Governance extends beyond transaction completion through the tracking of technical recommendations and risk mitigation measures, to capture identified synergies and monitor whether strategic objectives achieved. This is to ensure strategic transformation delivers sustained results, as the market increasingly rewards companies aligned with the energy transition (Savenkova, 2025).

From best practice principles, long-term tracking aligns with the critical success factors of advantaged portfolios: demonstrating resilience across diverse market conditions, sustainability through low emissions and regulatory compliance, predictability through stable operations and government support, and timeliness through market responsiveness and technology deployment (Harwood, 2025). This helps support portfolio transformation success to transcend individual project achievements and deliver strategic repositioning for sustained competitive advantage.

Discussion

Practical Framework Application Across Strategic Archetypes

Carbon pricing reshapes portfolio economics, driving companies toward distinct strategic archetypes while creating urgent rationalisation imperatives.

According to Savenkova, following three defined periods of asset growth, optimisation/digitalisation and decarbonisation, companies now face operation in a more disruptive environment. This shift is the culmination of distinct industry investment cycles over the past 15 years: a post-millennium era focused on growth to, a subsequent period of optimisation to reduce costs, and the recent imperative to transform and decarbonise (see Fig. 9).

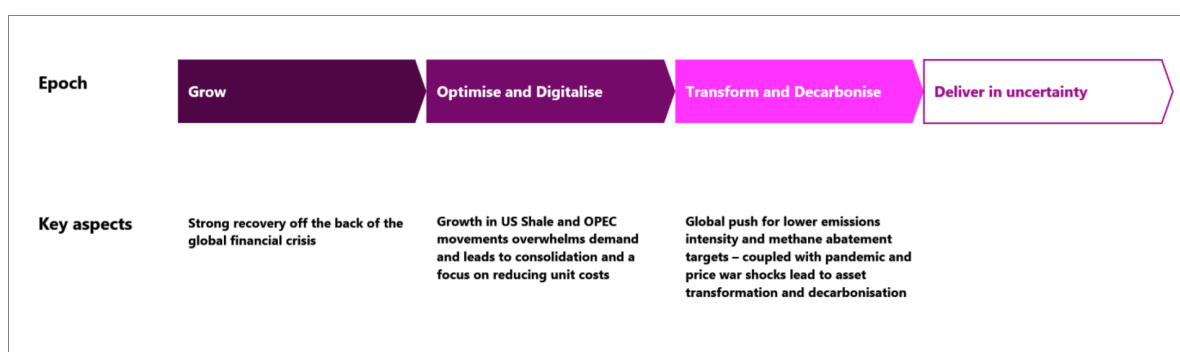


Figure 9—The historical oil and gas market has shifted through distinct epochs, with our likely moving into a new time of having to manage portfolios under significant uncertainty (Adapted from Savenkova (2025))

In the "Grow" epoch, large new markets like China were being brought into the global trading system, and the path to success was bringing resources to market as fast as possible. Following the oil price wars of the mid 2010s, the path to success was to "Optimise and Digitalise" to have the lowest unit cost. Then, following the landmark Paris Agreement at COP21 in 2015, which was operationalised around 2020 – driven by maturing investor frameworks like TCFD becoming mainstream and the specific, hard targets that emerged from the subsequent COP26 summit, and the declaration at COP28 in 2023 of the "Beginning of the End" of the Fossil Fuel Era – there was a clear path to success of managing Scope 1 / 2 emissions and

methane emissions in line with global pledges to be able to raise capital and stakeholder support ("Transform and Decarbonise").

Now, there is no clear path to success. Organisations have seen those who have doggedly pursued decarbonisation and failed. It is not sufficient to "decarbonise for decarbonisation's sake". This new epoch means new approaches to portfolio management that traditional M&A evaluations lack may be needed for organisations to navigate and deliver in uncertainty. Carbon pricing fundamentally alters portfolio economics, creating urgent portfolio rationalisation imperatives and driving companies toward distinct strategic archetypes.

Our framework utilises archetypes as a basis for strategy characterisation, recognising that building long-life, low-risk, high-margin portfolios represents the current industry objective. For resource specialists, this means high grading toward advantaged resources, shallow water locations, or conventional production techniques while implementing operational efficiency measures. For integrated energy players like Major EnergyCo, optimisation balances profitable hydrocarbon core assets with selective low-carbon investments offering genuine growth potential and portfolio resilience enhancement.

The evaluation process considers the asset operator and/or regional strategic positioning. In Australia, for example, LNG-weighted portfolios provide production longevity (declining <2% annually to 2034) and high cash flow margins (24% higher than global averages), while recognising higher emissions intensity (0.42 vs 0.26 tCO₂e/boe globally) requiring systematic mitigation strategies (Harwood, 2025). This evaluation enables objective prioritisation moving beyond purely financial or environmental decision-making to identify commercially attractive transactions meeting significant decarbonisation targets while maintaining strategic coherence.

Companies typically choose between three strategic archetypes: resource specialists (like Lundin Energy, now rolled up into Aker BP, focusing on profitable hydrocarbons with <3 kg/barrel emissions intensity), integrated energy players (like BP balancing core assets with low-carbon growth), or low-carbon pure plays (like Ørsted systematically divesting legacy assets).

While asset divestment transfers operational responsibility, we have evidence that acquiring operators are better positioned to optimise performance, often achieving superior emissions and economic outcomes through specialised focus and operational excellence.

A practical example demonstrates this principle: when a major international producer divested mature offshore assets to a regional specialist operator, the acquiring company doubled production from the same field while reducing offshore maintenance requirements by ~35% through focused operational improvements and specialised technical capabilities. This illustrates how strategic divestment can create value for all parties *provided investment/ divestment decisions are guided and aligned to an appropriate archetype/ strategy* which in turn relies on the self-recognition of ambition as matched to strengths, capabilities and motivations, as mentioned previously – the seller optimises its portfolio focus while the buyer leverages specialised expertise to enhance both operational and environmental performance.

Navigating these risks requires making clear strategic choices within a complex and diverging investment landscape, as illustrated in Fig. 10. Savenkova's (2025) analysis reveals no single winning formula for the energy transition, but demonstrates that a clear, well-executed corporate strategy is critical – the market can reward both focused hydrocarbon strategies and integrated diversification models while penalising unclear transformation efforts.

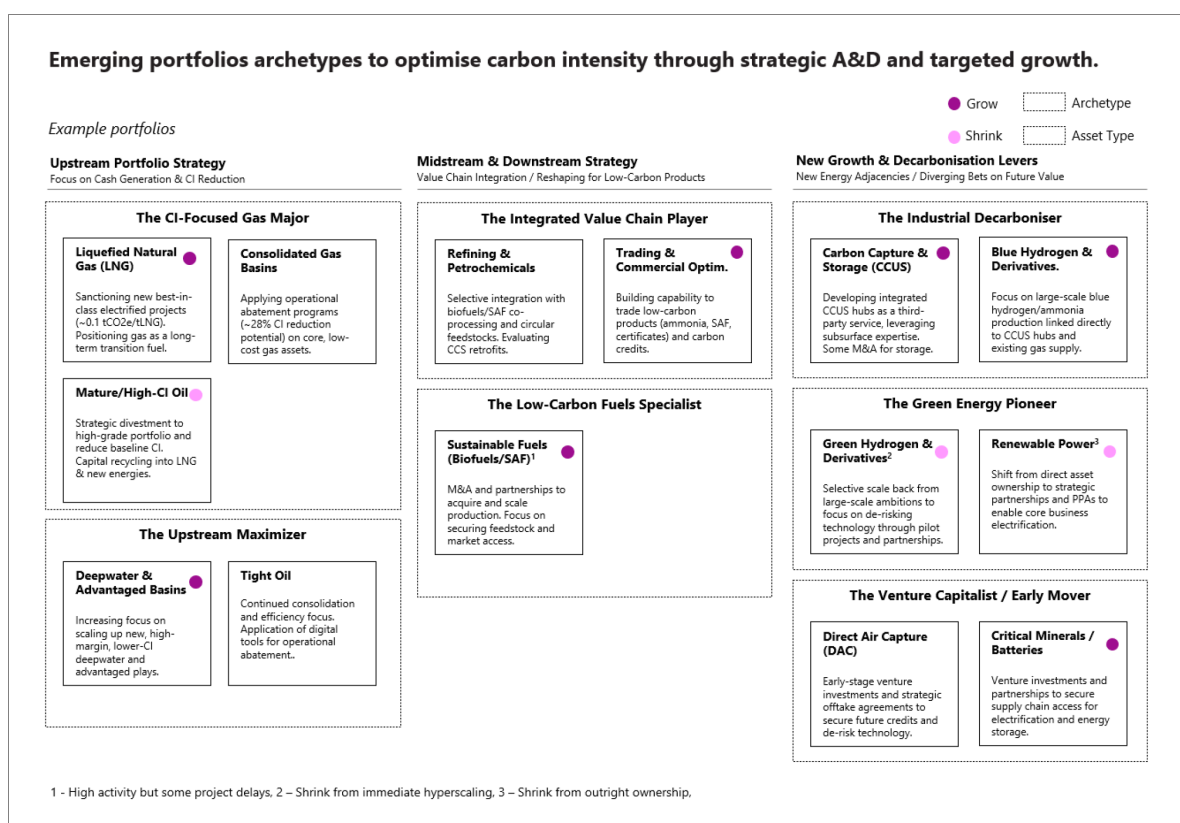


Figure 10—Emerging portfolios archetypes to optimise carbon intensity through strategic A&D and targeted growth.

When applying the framework, it is worth noting the need to adapt to diverse regional contexts and recognize that companies face different regulatory pressures, market conditions, and stakeholder expectations. This flexibility enables region-specific optimisation while maintaining systematic rigor (see Table 2).

Table 2—Regional operator / developer framework application examples

Regional Operator / Developer	Framework Application Example
Middle Eastern operators	Leverage the framework to achieve scale advantages and position as the world's lowest carbon intensity energy exporters, using advantages in a resource base and access to renewable power to dominate global markets on both cost and carbon performance.
European companies	Apply the methodology to meet stringent local regulations and demonstrate carbon leadership, using the governance structures to satisfy increasingly demanding investor and regulatory requirements while maintaining operational competitiveness.
Australian producers	Focus on unit carbon intensity optimisation to preserve long-term investment attractiveness, particularly for LNG exports to carbon-conscious Asian markets where environmental performance increasingly influences commercial terms.
North American operators	Emphasize operational efficiency gains that deliver both cost advantages and emissions reductions, aligning decarbonization with traditional performance metrics in competitive domestic markets
Asian energy importers	Balance energy security priorities with gradual transition pathways, using the risk management foundation to navigate evolving domestic policies while maintaining supply reliability

The application to Major EnergyCo demonstrates achievability of ambitious targets through coordinated A&D activities. The ~28% Carbon Intensity reduction achieved through portfolio reshaping, combined with operational optimisation programs, positions the company to exceed its 30% strategic target while maintaining commercial discipline and competitive positioning

Reflection on Archetype Implementation and Performance: Case Studies

The strategic choices made by companies like BP, Ørsted, and TotalEnergies in recent years provide compelling real-world examples of how the three-stage framework could be applied to managing portfolio transformation. [Savenkova's analysis \(2025\)](#) proposed, there is "no single winning formula" for the energy transition, but market rewards often hinge on "clear, well-executed corporate strategy" while "penalising unclear transformation efforts." We use the following miniature case studies to illustrate.

Ørsted: Missing Stage A Risk Management Foundation. Ørsted epitomizes a "low carbon pure-play" archetype as shown in [Fig. 10](#). Its systematic divestment of legacy assets and aggressive pivot to offshore wind created a world leader in renewable energy. While its current challenges could be interpreted by some as a failure of the low carbon strategy, an alternative view is its concentration of risk. Ørsted's bet was on a different world by a different timeline. Deeper application of the front-end "risk management" component of the framework may have identified a vulnerability to macroeconomic shifts through systematic baseline establishment and risk quantification. The subsequent surges in global inflation and interest rates combined with supply chain disruptions dramatically increased capital expenditure for offshore wind projects and eroded profitability of fixed-price power purchase agreements.

BP: Incomplete Stage B Value Communication and Stakeholder Alignment

BP's journey as an "integrated energy player" demonstrates the critical importance of the stakeholder alignment processes. Its ambitious strategy to balance reduced hydrocarbon footprint with significant low-carbon investments positioned it well within the integrated energy player archetype, similar to Major EnergyCo's approach in our case study.

However, BP's current challenges could be interpreted as an incomplete implementation of the stakeholder alignment methodology. The framework's approach to navigating competing stakeholder expectations between equity investors demanding near-term returns and debt providers focusing on longer-term ESG criteria may address similar market communication challenges. [Savenkova's \(2025\)](#) analysis showing an "inverse correlation" between majors' share prices and sustainability announcements suggests that BP struggled with portfolio communication. Despite achieving substantial operational progress to their strategy – a 38% oil and gas production reduction and 154% renewables growth – BP's 20% share price decline suggests insufficient application of carbon-adjusted valuation communication strategies that demonstrate both near-term financial benefits and long-term positioning advantages to different stakeholder groups.

TotalEnergies: Disciplined Stage A Foundation Enabling Selective Stage B Value Creation.

TotalEnergies' current success stems from rigorous application of the foundations of its archetype. Rather than pursuing aggressive transformation like BP or pure-play divestment like Ørsted, TotalEnergies made a calculated decision to leverage its existing advantaged gas assets as the foundation for integrated energy expansion.

TotalEnergies recognised its LNG assets as "advantaged" - high margins (resilience), access to clean electricity for future electrification (sustainability), established market relationships (predictability), and existing infrastructure (timeliness). Unlike BP's broad diversification approach, TotalEnergies chose a focused "integrated power" model that explicitly built upon its core LNG competencies rather than replacing them.

TotalEnergies' success reflects application of a strategic portfolio optimisation principle: it identified that its existing gas trading capabilities, LNG infrastructure, and customer relationships created natural synergies with renewable power integration – essentially applying a MACC analysis approach to find negative abatement cost opportunities through operational leverage rather than capital-intensive greenfield development.

Where BP struggled with stakeholder alignment because investors couldn't clearly value its diverse portfolio, TotalEnergies succeeded because its integrated power model demonstrated clear financial

logic: existing LNG customers needed reliable power supply with renewable firming capabilities, which TotalEnergies could provide through its established infrastructure and trading expertise. Its strategy was immediately comprehensible to stakeholders as natural portfolio extension rather than strategic pivot.

Unlike Ørsted's capital-intensive pure-play risk, TotalEnergies' avoided concentration risk by maintaining a diversified energy portfolio while capturing an upside from energy transition through operational leverage of existing advantages rather than wholesale business model transformation.

This more disciplined, framework-aligned approach may explain the current superior market performance – it achieved transformation benefits while maintaining the strategic coherence and risk management that the framework emphasizes.

Operational Teams play a direct part in portfolio shaping with key day-to-day actions and through awareness of the broader organisational context.

Operational teams face increasing regulatory mandates for emissions reductions, with many jurisdictions requiring 30-40% carbon intensity improvements by 2030. While strategic portfolio reshaping provides the broad pathway at a corporate level, operators need approaches to achieve these targets through practical operational improvements and targeted asset and capital project optimisation (see [Table 3](#)).

Table 3—Operational implementation guidelines for key roles in a hydrocarbon operating company

Operating Company Role	Direct Part to Play in Carbon Intensity Reduction
Asset Managers	Use carbon intensity baselines to identify highest-impact operational improvements
Engineers	Integrate carbon intensity and methane reduction into routine maintenance and process optimisation alongside energy optimisation.
Operations Teams	Track carbon intensity alongside traditional KPIs (uptime, costs, production)
Project Teams	Include carbon intensity impact assessment in major CAPEX decisions

Conclusion

Strategic Portfolio Transformation provides the basis for Competitive Advantage in this new Epoch in the Energy Transition

This framework provides a comprehensive methodology for embedding Carbon Intensity with recognition as a core risk and value driver throughout the complete A&D lifecycle.

Rather than treating carbon intensity as an external constraint, the framework demonstrates systematic integration within standard investment processes. Traditional M&A evaluation lacks systematic integration of Carbon Intensity as a core performance metric, leaving companies exposed to transition risks. The three-stage methodology transforms this by embedding Carbon Intensity-weighted screening, carbon-adjusted valuation, and portfolio-level governance, converting decarbonization from regulatory compliance into strategic value creation.

The framework provides replicable processes applicable across different asset portfolios, strategic contexts, and market environments. Whether companies position as resource specialists focused on advantaged hydrocarbons, integrated energy players balancing traditional and low-carbon assets, or low-carbon pure plays systematically transitioning portfolios, the methodology enables effective execution while maintaining commercial discipline and competitive advantage.

Our success with Major EnergyCo in implementing a plan towards a ~30% Carbon Intensity reduction with \$86 million positive NPV demonstrates practical applicability for energy incumbents facing similar transformation pressures. The framework's integration of quantitative risk assessment, due diligence, carbon-adjusted valuation (incorporating \$50-70/tCO₂e pricing scenarios), and advanced portfolio governance (PMO tracking with AI-driven optimisation) provides proven methodologies that deliver both environmental and financial results.

As market dynamics increasingly reward strategic clarity and execution capability, systematic portfolio transformation becomes essential for long-term competitiveness. Through this framework, energy companies can identify and execute the most commercially attractive transactions to simultaneously achieve ambitious decarbonisation targets and commercial objectives, positioning them to lead rather than merely adapt to transition pressures. The methodology transforms portfolio reshaping from ad-hoc opportunism into strategic capability essential for long-term value creation and energy transition leadership in an evolving energy landscape.

Abbreviations used in paper

A&D	acquisition and divestment
AI	artificial intelligence
API	American Petroleum Institute (as used in "API gravity")
CAPEX	capital expenditure
CCGT	combined cycle gas turbine
CoLab	Collaborative Innovation (as used in "CoLab workshops", a Wood brandname)
DD	due diligence
ESG	environmental, social and governance
KPI	key performance indicator
LCA	lifecycle assessment
M&A	mergers and acquisitions
MACC	marginal abatement cost curve. A graphical tool that ranks emission reduction opportunities by cost-effectiveness, showing the cost per unit of emissions avoided for different abatement measures. "Negative cost" options on the left are measures that save money while reducing emissions, such as energy efficiency improvements. Towards the right are increasingly expensive options, such as carbon capture technologies.
NPV	net present value. NPV is a financial metric that calculates the current worth of a future stream of cash flows by discounting back to today's dollars, less the initial investment cost. This is used because money today is worth more than the same amount in the future (due to inflation, risk, and opportunity cost). NPV is used for investment decision-making, project evaluation, and comparing mutually exclusive alternatives because it directly measures value creation in today's dollars.
OPEX	operational expenditure
PEP	project execution plan
PMO	portfolio management office
SME	subject matter expert
TCFD	Task Force on Climate-related Financial Disclosures

References

- Carydias, P. & Flowers, S. (2024). Charting a Path to Low-Carbon Leadership in the GCC: Unlocking up to 90% Emissions Reductions in Greenfield LNG. SPE-222347-MS, ADIPEC 2024.
- Carydias, P. & Nielsen, P. (2023). Towards 2030: Transforming Asset Emissions in Upstream Production Operations. SPE-216006-MS, ADIPEC 2023.
- Granskog, A., Patel, M., Gupta, R., & Helmcke, S. (2025, April). How incumbents can succeed in climate-driven growth investments. McKinsey & Company.
- Harwood, A. (2025). Corporate strategies for a new energy transition outlook: advantage Australia? *Australian Energy Producers Journal*, **65**, EP24247.
- Savenkova, O. (2025). Navigating and leading corporate strategy amid energy transition uncertainty. *Australian Energy Producers Journal*, **65**, EP24058.